

P3 - Paneling, Arrays and Attractor Façade

ARC 3133-001 / Architectural Visualization Methods 1 / Fall 2026 - Full Term

Revision: 2026-09-22. Instructor review draft. Dates and weights are proposed; review before publication.

3.1: module, authored material, ground plane and three-point lighting. 3.2: arrays with library or student-authored materials. 3.3: attractor-driven panel façade, shown in elevation and isometric.

3.1 - Descriptive geometry module - material and three-point lighting

Introduced: Week 4 - Sep 15 / **Due:** Week 9 - Oct 20, 11:59 PM Eastern Time / **Weight:** 7%

Description

Construct a panel from a built precedent using explicit point positions and ordered faces. Verify closure and tiling, create one material for the module, and present it resting on a ground plane with a three-point lighting setup. This authored module becomes the unchanged element in 3.2 and the parameter-driven element in 3.3.

Rules & Constraints

- Choose a simple panel from a built precedent that tiles edge to edge in a 2 x 2 grid. Credit the building, architect and source.
- Define parameters first and explain each. Include a shape control such as aperture, fold depth or corner offset, beyond overall dimensions.
- Build an editable, closed mesh from explicit point and face lists. Calculate point positions from the parameters.
- State the ranges where the module closes and tiles. Diagram low, middle and high values of a shape parameter.
- Use one 17 x 11 inch sheet with a studio render of your module resting on a ground plane. Create one material yourself and apply it to the module. Set up three-point lighting with a key light, fill light and rim/back light. Use a simple neutral ground material and no entourage.
- Beside the render, show point and face lists, a parameter table and the vertex order of one face. Explain normals, closure and reversed face winding.
- Show a lighting diagram identifying the camera, ground plane and three lights. Record each light's type, position, size and power, and the material settings needed to reproduce the render. Three-point lighting describes the three lighting roles; it does not require three Point-type lights. Area lights are suitable.

Parameters

- Width, depth and height/thickness in stated model units, with a local origin and X/Y/Z axes.
- At least one shape control beyond overall size: aperture, fold depth, corner offset or an equivalent geometric feature. Explain how this value changes specific point coordinates.
- Valid ranges and dependencies for every control. State the limits within which faces remain valid, the solid closes and neighboring panels meet.
- Your authored module material: base color, roughness and any other active settings. Three lights: key, fill and rim/back, with type, position, size and power recorded. Use a neutral ground plane; keep its material separate from the one authored module material.

Required Studies

- **Precedent and construction.** Credit the building, architect and image source. Diagram the panel boundary, numbered vertices, connecting edges and ordered face lists. Show the vertex order of one face and explain how reversing it changes its normal.

- **Descriptive views.** Use coordinated plan/front/side views and an axonometric construction diagram to show how the point coordinates generate the form. Dimension the boundary and the shape-controlled feature; distinguish construction lines from final edges.
- **Three parameter settings.** Show low, middle and high values of one shape parameter from the same graph. Label all settings. Verify closure and show how the selected module tiles edge to edge in a 2 x 2 grid.
- **Studio render and lighting diagram.** Place the module on a ground plane so contact and cast shadows explain its position. The key light establishes form, the fill controls shadow contrast, and the rim/back light separates the edges from the background. Show the three lights and camera in a small diagram; record the material and light settings. No entourage.

Workflow

1. Select a simple built panel with a repeatable boundary. Identify its point/edge/face relationships before creating the graph.
2. Define named inputs, calculate the vertices and assemble the face lists in consistent order. Build the module yourself from these elements.
3. Check closure, normals and shared tile boundaries at the low/middle/high parameter settings. Resolve gaps, reversed faces or self-intersections.
4. Create the module material yourself. Place the module on a ground plane and build the key/fill/rim lighting setup. Test each light separately, then balance the combined result. Compose the sheet with the construction evidence, render and lighting diagram.

Deliverables

- **One 17 x 11 inch landscape module sheet:** Lastname_Firstname_P3a.pdf. Include the credited precedent, construction and descriptive views, point/face lists, parameter table, three shape settings, 2 x 2 tiling check and studio render with settings. Include the three-point lighting diagram and identify the authored material.
- **Editable graph/source:** .blend, or an already approved .gh/.3dm route. Include the material/light scene used for the render and package referenced assets. Keep native node names visible and explain your groups with Frames.
- Submit files in Lastname_Firstname_P3a/. Save the chosen parameter setting and panel origin so the exact same module can be reused in P3b. No physical print is required for this milestone.

Evaluation

- **Geometric reasoning:** point and face lists, ordered vertices, normals and closure explain the constructed solid.
- **Control and validity:** a meaningful shape input produces the three documented alternatives inside a tested range.
- **Tiling:** the selected panel meets its neighbors edge to edge in the 2 x 2 check.
- **Material and presentation:** the student-authored material and key/fill/rim lighting reveal the geometry; the module visibly rests on a ground plane; the lighting diagram and recorded settings match the editable scene.

Before You Submit

- [] Built precedent credited; panel authored from calculated points and ordered faces.
- [] One shape control beyond dimensions, tested at low/middle/high values.
- [] Closure, normals and edge-to-edge 2 x 2 tiling demonstrated.
- [] Point/face lists, descriptive views and parameter values are readable.
- [] One student-authored module material; ground plane; key, fill and rim/back lights with a diagram and settings; no entourage.

3.2 - Arrays

Introduced: Week 6 - Sep 29 / **Due:** Week 10 - Oct 27, 11:59 PM Eastern Time / **Weight:** 7%

Description

Use your P3a panel to explain repetition in one, two and three dimensions. Keep the module geometry unchanged and compute its placement from indices, counts and spacing. Show both the arrangement and the rule that generates it. Attractors and other array families are outside this brief's current scope.

Rules & Constraints

- Array your P3a module across exactly two 17 x 11 inch sheets, submitted together as one assignment.
- Sheet 1 - 1D, 2D and 3D arrays: a linear array, a nested grid and an XYZ cube array. Label count, spacing and index order; show the predicted total for each.
- Sheet 2 - parameters and index logic: a second count/spacing setting for each 1D, 2D and 3D array, a row/column/layer diagram, and the conditional selection before and after its rule applies.
- Current brief scope: 1D, 2D and 3D arrays. Hexagonal, radial and curve arrays and attractors are reserved for later work.
- Keep the module geometry unchanged; vary its placement. Expose count and spacing in the graph.
- Build repetition from basic nodes and begin with a count you can inspect; there is no minimum element count.
- Include one conditional selection on either sheet. State its rule (for example, indices 3-5 lift by 0.5) and show the array before and after it applies.
- Use materials from the instructor-provided library or create your own. Record which materials you used and any edits; package required textures. Material choice must keep array spacing, depth and repetition readable. The library will be posted before this assignment is due; creating your own remains an equally valid route.

Parameters

- Integer counts N_x , N_y , N_z and positive step distances dx , dy , dz . Step distance measures between corresponding module origins; it is not necessarily the clear gap between edges.
- Base point P_0 , array axes and one shared module orientation. Label the local origin used to place each copy.
- A conditional selection rule and its effect, such as lifting a named index range. Expose the selection limits and displacement without changing the module itself.
- Material selection and any edited settings. Use the instructor-provided library or your own materials; identify the source and package textures. There is no separate requirement to author a new material for 3.2.

Required Studies

- **1D - row.** Vary one index: $P(i) = P_0 + i \cdot dx \cdot X$. Label the first few indices, count and spacing. Predict the number of copies before evaluating the graph.
- **2D - grid.** Use independent row/column indices: $P(i, j) = P_0 + i \cdot dx \cdot X + j \cdot dy \cdot Y$. Predict $N_x \cdot N_y$ copies. Show the row/column structure and distinguish it from manually duplicating a row.
- **3D - spatial array.** Add an independent layer index: $P(i, j, k) = P_0 + i \cdot dx \cdot X + j \cdot dy \cdot Y + k \cdot dz \cdot Z$. Predict $N_x \cdot N_y \cdot N_z$ copies. For equal counts N , the total is N cubed; 3, 4 and 5 per axis give 27, 64 and 125 copies. Show an axonometric view plus a layer view that makes the Z repetition visible.
- **Controlled comparisons.** For each array, change count and/or spacing using the same graph and record both settings and totals. Keep the panel shape fixed. Begin with inspectable counts; no large minimum element count is required.
- **One conditional.** State an index-based rule and show before/after views. Identify the affected copies and the displacement. Ensure the selection leaves the other copies unchanged.

Workflow

1. Reference the selected P3a module, preserving its dimensions, shape controls and origin. Separate panel generation from placement.
2. Build a small 1D row, then nested 2D and 3D repetitions from basic nodes. Use $i = 0..N_x-1$, $j = 0..N_y-1$, $k = 0..N_z-1$ and define X/Y/Z as unit vectors of the stated axes.
3. Predict and inspect counts at each stage. Explain indices and row/layer organization. Distinguish Grasshopper lists/trees from Blender fields or Repeat Zones; use the appropriate data structure for your software.
4. Test the second count/spacing settings and the conditional selection. Check for duplicate positions and unintended collisions, then compose the two sheets. Apply library materials or your own materials consistently across the comparison views.

Deliverables

- **Exactly two 17 x 11 inch landscape sheets in one PDF:** Lastname_Firstname_P3b.pdf. They are one assignment, submitted together.
- **Sheet 1 - 1D, 2D and 3D.** Show the row, nested grid and spatial array with their placement rules, labeled axes, count/spacing values and predicted totals. Include the 3D layer view and a small reference image of the unchanged P3a panel.
- **Sheet 2 - Parameters and index logic.** Show the second setting for each array with values and totals, a diagram of row/column/layer organization, and the conditional before/after comparison with its rule.
- **Editable graph/source** in Lastname_Firstname_P3b/, with labeled inputs and an output for each array. Internalize Grasshopper references or package equivalent dependencies. Save/record the settings used on both sheets so they can be reproduced. No physical print is required. Include a short material-source note and all referenced textures.

Evaluation

- **Array logic:** row, grid and spatial array use one, two and three independent directions; predicted and actual totals agree.
- **Data control:** index order and row/layer organization are understandable; the conditional affects exactly the intended copies.
- **Parametric behavior:** count and spacing changes recompute the arrangement while preserving the P3a module.
- **Design and communication:** views reveal spacing and dimensional structure; captions, diagrams and file organization follow the Graphic Standards Manual.

Before You Submit

- [] Same P3a panel used unchanged in all arrangements.
- [] A row, a grid and a true spatial array are clearly distinguished.
- [] Axes, count, spacing, index order and predicted totals are shown for each.
- [] A second setting for every array and one conditional before/after comparison are documented.
- [] Two sheets submitted together; the source reopens and reproduces the documented outputs.
- [] Material library selections or student-authored materials identified; referenced textures packaged.

3.3 - Attractor-driven panel façade

Introduced: Week 11 - Nov 3 / **Due:** Week 13 - Nov 17, 11:59 PM Eastern Time / **Weight:** 7%

Preparation checks: Week 12 - Nov 10: Bring a uniform façade, the sampled attractor field, and a first test of one driven panel parameter.

Description

Turn your panel array into a responsive façade. Measure an attractor field across the façade and use its values to change one geometric parameter in the panel you built in 3.1. The elevation explains the overall composition; the isometric explains the three-dimensional result.

Rules & Constraints

- Use your 3.1 panel and the 2D array logic from 3.2 to build a façade. Sample distance to a point or curve attractor at each panel location.
- Use the attractor data to drive exactly one geometric parameter of the panel, such as aperture, fold depth or corner offset. Recompute panel geometry from that parameter. Color alone or moving unchanged panels does not satisfy this requirement.
- Diagram distance measurement, normalization/remapping, clamping and the chosen parameter range. Keep all other panel geometry inputs fixed for the comparison. Stay inside the valid range tested in 3.1.
- Compare a uniform baseline with the attractor-driven façade. Show one additional attractor position using the same graph and parameter mapping.
- Present one 17 x 11 inch composition with an orthographic façade elevation and a matching isometric view of the same final façade. Make the attractor location/curve and its influence readable in an overlay or accompanying field diagram with a legend.
- Submit a print-ready PDF and the editable graph. These are views of a 3D panel system; a standalone 2D line-field drawing is no longer the assignment. Pen plotting is not required.

Parameters

- Façade row/column counts, panel spacing and base plane/origin. Use a stable grid so the comparison isolates changes to panel form.
- Attractor point position or curve, measured-distance input range, remapping/falloff and clamping.
- One selected geometric panel parameter, its units and minimum/maximum values. Keep the other shape parameters fixed; test validity at both extremes.

Required Studies

- **Baseline.** Show a small uniform façade with the selected panel parameter fixed. Label the constant value.
- **Field to form.** Diagram sample location to distance to normalized/clamped value to panel parameter. Show the field and legend beside the geometry so the relationship is traceable.
- **Controlled variation.** Move the attractor once and compare the result while keeping the grid, range and remaining inputs unchanged. Do not rebuild the graph for each study.
- **Final composition.** Give the elevation and matching isometric the most space. Show the attractor explicitly and label the selected parameter, limits and final settings. Use the Graphic Standards Manual.

Workflow

1. Reference your own panel definition and create a small façade grid. Confirm the uniform baseline first.
2. Measure distance at each panel location. Map and clamp the values into the chosen panel parameter range, then generate the individual panels.

3. Check the near and far limits and several intermediate panels. Resolve gaps, self-intersections or invalid faces before increasing the array count.
4. Capture the two attractor settings and compose the elevation, isometric and explanatory diagrams from the saved final state.

Deliverables

- **One 17 x 11 inch landscape PDF:** Lastname_Firstname_P3c.pdf. Include the final façade elevation and isometric, the attractor/field legend, the mapping diagram, the baseline and one additional attractor setting.
- **Editable source** in Lastname_Firstname_P3c/: save the panel definition, array, attractor and named mapping controls; internalize or package all dependencies and materials. Record the settings used for the final views.
- Prepare the composition for normal printing at its stated size. No pen-plotter output or physical façade model is required.

Evaluation

- **Attractor logic:** a measured field drives one actual panel geometry parameter through an explainable mapping.
- **Continuity:** the work develops the student's 3.1 panel and 3.2 array; extreme values remain geometrically valid.
- **Experimentation:** baseline and changed-attractor studies isolate the effect of the chosen control.
- **Design and communication:** elevation, isometric, field legend and captions agree and clearly explain the façade composition.

Before You Submit

- [] Own panel and array reused; one geometric panel parameter responds to the attractor.
- [] Input/output ranges, units, clamping and attractor position are documented.
- [] Uniform baseline and one additional attractor setting are shown.
- [] Final elevation and isometric show the same façade; field influence is legible.
- [] Print-ready composition and packaged source reproduce the final result.

Shared standards: Course software, fabrication, attribution and AI policies apply. Submit through the channel announced in class/Canvas. Test the packaged source from a clean folder. Percentages follow the instructor review draft. This handout uses US Letter; student sheets are 17 x 11 inches.

Midterm grade: The midterm grade accumulates the Graphic Standards Manual, CSG sheet and print, module (3.1) and arrays (3.2). These assignments account for 35% of the final course grade. Report the accumulated result as a percentage of those 35 possible course points. The midterm adds no separate assessment weight.